

Software Requirements Specification (SRS)

Hostel Management System

1. Introduction

1.1 Purpose

This document provides a detailed description of the requirements for the Hostel Management System. It is intended for developers, designers, project managers, and stakeholders.

1.2 Scope

The system will digitize hostel operations including student management, room allocation, food services, payments, and complaints.

1.3 Definitions

- Admin: System manager
- Student: Hostel resident
- SRS: Software Requirements Specification

2. Overall Description

2.1 Product Perspective

A mobile/web-based application using modern technologies.

2.2 Product Functions

- Registration & Login
- Room Management
- Meal Management
- Payment System
- Complaint System
- Admin Dashboard

2.3 User Classes

- Admin

- Student
- Manager (optional)

2.4 Operating Environment

- Android / Web
- Internet required

3. Functional Requirements

3.1 Authentication

- User registration
- Login/Logout

3.2 Room Management

- Add/update/delete rooms
- Room request system
- Approval by admin

3.3 Food System

- View menu
- Meal subscription

3.4 Payment System

- Fee payment
- Payment history

3.5 Complaint System

- Submit complaint
- Track status

3.6 Admin Panel

- Manage users
- Manage rooms
- Monitor payments

4. Non-Functional Requirements

4.1 Performance

- Response time under 3 seconds

4.2 Security

- Authentication required
- Secure data storage

4.3 Usability

- User-friendly UI

4.4 Reliability

- High availability

4.5 Scalability

- Supports many users

5. System Features

Login

Input: Email, Password

Output: Dashboard

Room Booking

Input: Request

Output: Approval Status

Payment

Input: Amount

Output: Confirmation

6. External Interface Requirements

UI

- Mobile responsive

Software

- Firebase

Communication

- Internet

7. Technology Stack

Frontend

- Flutter / Android Studio

Backend

- Firebase Authentication
- Firebase Firestore

Programming Languages

- Dart / Java / Kotlin

Tools

- Figma (UI Design)
- Git (Version Control)

8. Advanced Features

- Notification System
- Online Payment (bKash/Nagad)
- Smart Room Allocation
- Attendance Tracking
- Digital Invoice
- Maintenance Management
- Analytics Dashboard
- Meal Subscription
- Role-Based Access
- Profile Verification

- Security (OTP, Encryption)
- Multi-language Support
- Inventory Management
- Chat System
- Google Maps Integration

9. Database Overview

Collections:

- Users
- Rooms
- Payments
- Complaints
- Meals

10. Future Enhancements

- AI Features
- Biometric Login
- SMS Notification

11. Conclusion

This document describes a complete and professional Hostel Management System. Following this will ensure a scalable and efficient system.